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CUSTOMER SHOPPING BEHAVIOR SIMULATOR

An Empirical First-Order Markov Chain Model of E-Commerce Sessions

Research Paper

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Abstract

This paper presents an interpretable simulator for e-commerce shopping sessions using a first-order, discrete-time Markov chain. Raw event logs are converted into visitor sessions, mapped to four behavioral states - BROWSING, CART, PURCHASED, and EXITED - and used to estimate transition probabilities from a chronological training partition. The fitted model generates synthetic journeys that are compared with a held-out partition. On the supplied Retailrocket events file, a reproducible run sampled 10,000 visitor identifiers and retained 18,408 event rows, forming 12,240 sessions. The chronological split contained 9,792 training and 2,448 holdout sessions. With 1,000 simulations and random seed 42, the holdout purchase proportion was 0.6127% (15 of 2,448 sessions), while the simulation purchase proportion was 0.6000% (6 of 1,000 journeys); the absolute difference was 0.0127 percentage points. This close aggregate match is descriptive evidence for this sample and run, not a claim of individual-level prediction or generalizable accuracy. The work emphasizes transparent assumptions, reproducibility, and the distinction between observed and simulated outcomes.

Keywords: e-commerce analytics; clickstream; customer journey; Markov chain; absorbing states; sessionization; stochastic simulation.

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1. Introduction

Online stores record sequences of interactions such as item views, cart additions, and transactions. Aggregate event counts describe how frequently these actions occur, but not how sessions move from one action category to another. This project addresses that gap with a compact state-transition model that represents a shopping journey as a sequence of behavioral states.

The aim is to build a reproducible analysis pipeline: validate event records, group them into sessions, estimate transition probabilities, simulate journeys, and compare simulated terminal outcomes with sessions held out from model estimation. The model is intentionally interpretable. It provides a baseline for describing sequence behavior and for later comparison with more complex methods.

1.1. Research question

Can a first-order Markov chain estimated from a chronological training subset reproduce the broad purchase and exit proportions observed in a held-out subset of the sampled visitor sessions?

1.2. Contributions

The project contributes an end-to-end implementation for raw e-commerce events, an explicit session and state definition, a row-normalized transition matrix, reproducible journey simulation, holdout comparisons, and exported tables and figures. The work is an applied probabilistic simulation study; it does not implement a neural network.

2. Background and related concepts

A discrete-time Markov chain represents a sequence of states in which the probability of the next state depends on the current state [2]. In this study, each step is a transition between mapped shopping stages, rather than a fixed unit of clock time. The assumption is useful for a small, transparent baseline, but it discards parts of the visitor's prior history.

Prior work has applied Markov chains to clickstream analysis and online customer journeys. Scholz describes first- and higher-order Markov approaches for clickstream data, including an online-store example [3]. Research on browsing-path prediction also cautions that a first-order Markov model can perform poorly when path history matters [4]. These findings motivate treating the present model as an interpretable baseline and keeping its predictive claims modest.

Purchase and exit are modeled as absorbing states: once a simulated path reaches either terminal state, it stops. The two transient states are BROWSING and CART. The transition probabilities are estimated by relative frequency, so each observed transient row is normalized by its outgoing transition count. No smoothing is applied.

3. Problem formulation

Let X_t be the behavioral state at step t, with state space $S = \{\text{BROWSING}, \text{CART}, \text{PURCHASED}, \text{EXITED}\}$. A first-order model assumes $P(X_{t+1}=j \mid X_t=i, X_{t-1}, \ldots, X_0) = P(X_{t+1}=j \mid X_t=i) = p_{ij}$. The transition matrix $P = [p_{ij}]$ is estimated from adjacent state pairs in training sessions.

A journey begins in BROWSING by default. From each transient state, the next state is sampled from its estimated row in P. Simulation ends at PURCHASED, EXITED, or a configured step limit. Separately, the observed held-out session outcome is its final modeled state under the session-construction rules.

4. Dataset and preparation

The input is the Retailrocket E-commerce Dataset behavior log, downloaded from Kaggle [1]. The published collection describes event interactions from a real e-commerce website and includes view, add-to-cart, and transaction events. The local events.csv used here contains 2,756,101 data rows and 94,237,913 bytes. Its columns are timestamp, visitorid, event, itemid, and transactionid. All identifiers are treated as opaque keys; the simulator uses timestamp, visitor ID, and event type for this analysis.

Quantity	Value
Rows in supplied file	2,756,101
Distinct visitors sampled	10,000
Rows retained in sampled analysis	18,408
Sessions formed	12,240
Train / holdout sessions	9,792 / 2,448
Retained event types	17,866 views; 436 add-to-cart; 106 transactions

The visitor sample is a subset of visitor identifiers, not a random sample of 10,000 event rows. All available rows for selected visitors are retained, which is why 10,000 sampled visitors produced 18,408 event records. In this run, 18,408 of 18,408 read rows passed the implemented cleaning checks; zero were excluded.

4.1. Session construction and state mapping

Timestamps are interpreted as Unix milliseconds in UTC. Events are ordered within visitor. A new session begins when the gap between consecutive events is strictly greater than 30 minutes. This inactivity threshold is a configurable operational definition, not a ground-truth session label. Event types map as follows: view to BROWSING; addtocart to CART; transaction to PURCHASED. The sequence is truncated at the first transaction. Sessions without a transaction receive an inferred EXITED terminal state. These rules simplify observed clickstream activity into a small finite state space.

5. Methodology

5.1. Chronological partition

Sessions are ordered by session start time and split chronologically: the first 80% are used for transition estimation and the remaining 20% are held out. This preserves complete sessions in one partition and makes the comparison later in time than the training data. In this run, there were 9,792 training sessions and 2,448 holdout sessions.

5.2. Transition estimation

For each adjacent pair of states (i,j) in training sessions, a transition count C_ij is accumulated. For a transient state with observed outgoing transitions, its probability is estimated as p_ij = C_ij / sum_j C_ij. The terminal rows are defined as absorbing: PURCHASED transitions to PURCHASED with probability 1, and EXITED transitions to EXITED with probability 1. No pseudo-counts or smoothing are added; consequently, estimates for rare actions can be sensitive to sample size.

5.3. Simulation and absorbing outcomes

The simulator uses a seeded NumPy random generator to draw the next state from the current row of the transition matrix. Each simulated path starts in BROWSING by default and stops at a terminal state or at the step limit. With transient-state block Q and absorption block R, theoretical eventual absorption probabilities can be computed as B = (I - Q)^(-1) R when the inverse exists. These probabilities are properties of the fitted chain, not guaranteed future conversion rates.

5.4. Evaluation measures

Terminal outcome proportions are compared between holdout sessions and simulated journeys. For outcome k, absolute error is |p_observed(k) - p_simulated(k)|. Transition frequencies are also compared descriptively. This evaluation does not constitute a calibrated forecasting benchmark: simulation outcomes have Monte Carlo variation, and results depend on the visitor sample, session threshold, state mapping, chronological split, seed, and number of simulated journeys.

6. Implementation

The application is implemented in Python. Pandas is used for tabular event processing, NumPy for seeded sampling and numerical calculations, Matplotlib for static figures, and Streamlit for an interactive dashboard. The command-line workflow and dashboard share the project pipeline. Outputs include data-quality summaries, event counts, session summaries, transition counts and probabilities, validation tables, example simulated paths, and charts.

The implementation also includes a synthetic development mode for checking behavior without confusing generated example data with the supplied real event file. Tests cover core data-processing and model behavior. The dashboard labels sampled input and simulated paths so generated outcomes are not presented as observed facts.

7. Experimental setup

Setting	Value
Input	Retailrocket events.csv
Visitor sample	10,000 identifiers
Session inactivity boundary	30 minutes
Partition	Chronological 80% train / 20% holdout
Simulation count	1,000 journeys
Random seed	42
Starting state	BROWSING
Maximum simulation steps	Configured limit; no run reached it

The selected run retained 18,408 rows and formed 12,240 sessions. No rows were excluded by the configured quality checks. The selected visitor sample and run settings are fixed for the results reported below; changing them may change the matrix and the observed/simulated comparison.

8. Results

8.1. Estimated transition probabilities

Table 3 reports the row-normalized training matrix. From BROWSING, the estimated next-state probabilities were 30.93% for another BROWSING step, 1.55% for CART, 0.15% for PURCHASED, and 67.37% for EXITED. From CART, the estimates were 36.51% to BROWSING, 15.56% to CART, 13.02% to PURCHASED, and 34.92% to EXITED. The small estimated direct BROWSING-to-PURCHASED probability should be read in light of event abstraction and sparse transaction transitions.

From / To	BROWSING	CART	PURCHASED	EXITED
BROWSING	0.3093	0.0155	0.0015	0.6737
CART	0.3651	0.1556	0.1302	0.3492
PURCHASED	0.0000	0.0000	1.0000	0.0000
EXITED	0.0000	0.0000	0.0000	1.0000

Table 3. Estimated transition matrix from the 9,792 training sessions. Terminal states are absorbing by model definition.

8.2. Held-out and simulated terminal outcomes

Outcome	Observed holdout	Simulated	Absolute difference
PURCHASED	0.6127% (15/2,448)	0.6000% (6/1,000)	0.0127 percentage points
EXITED	99.3873% (2,433/2,448)	99.4000% (994/1,000)	0.0127 percentage points
STEP_LIMIT	0.0000% (0/2,448)	0.0000% (0/1,000)	0.0000 percentage points

Table 4. Holdout outcomes compared with 1,000 simulated journeys. Absolute differences are expressed in percentage points.

The purchase proportions differ by approximately 0.0127 percentage points in this run. This is a close aggregate match on the chosen holdout, but the quantities are based on only 15 observed purchases and six simulated purchases. A small absolute difference here should not be interpreted as proof of strong predictive performance. The exit proportions are correspondingly similar because exit is the dominant terminal outcome.

8.3. Transition visualization

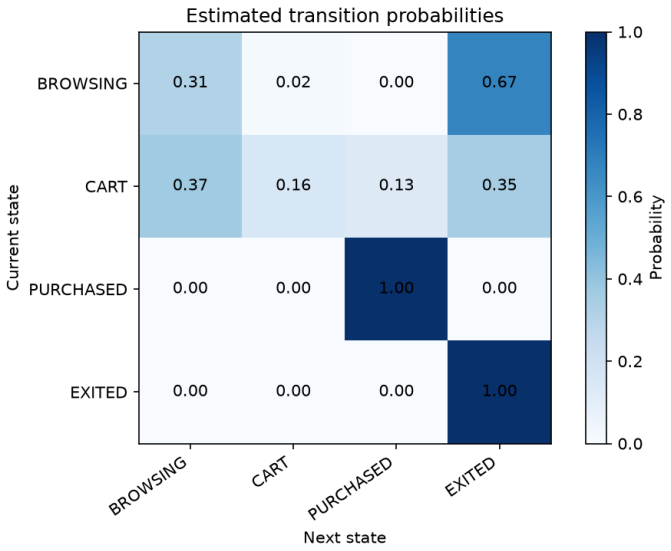


Figure 1. Training transition probabilities across the four modeled behavioral states.

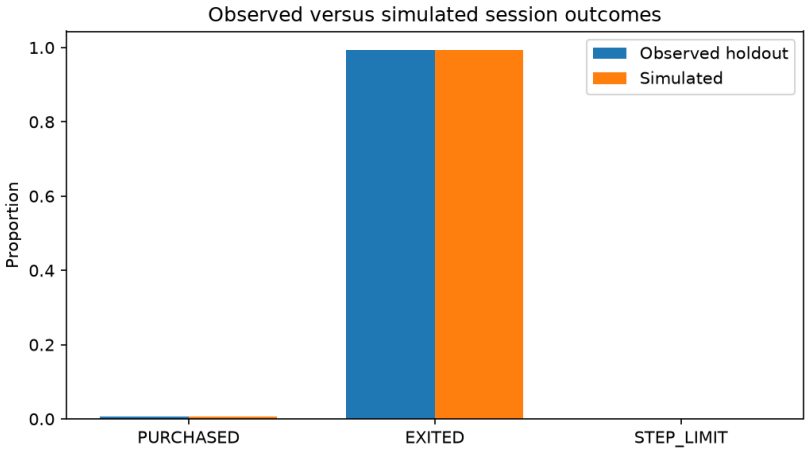


Figure 2. Observed holdout and simulated terminal outcome proportions. The plotted values are sample-specific.

9. Discussion

The transition matrix makes the model's assumptions visible: browsing is frequently followed by an inferred exit, while cart sessions show several possible next states. The simulation reproduces the overall outcome mix closely for the selected holdout and seed. Because exits dominate and purchases are rare, outcome agreement is driven largely by the high exit proportion. Evaluating purchase behavior more robustly requires larger samples, repeated seeds, uncertainty intervals, and comparisons across temporal windows.

This approach is useful as an educational and analytical baseline because it exposes the link between observed transition counts and simulated paths. It is not a recommendation engine and does not estimate product-level preference, revenue, or individual intent. The dashboard should be interpreted as an exploratory interface for the fitted model.

10. Limitations and threats to validity

First, the four-state abstraction compresses detailed behavior and omits item identity, product category, price, promotions, device, and other context. Second, a 30-minute gap is a chosen session rule and can split or combine behavior differently from a platform's own session definition. Third, EXITED is inferred when no transaction is observed; it is not directly recorded in the source events.

Fourth, the first-order Markov assumption makes the next-step probability depend only on the current state. Real shopping behavior may depend on prior steps, time, and visitor attributes. Fifth, no smoothing is used, so low-frequency transitions are noisy and some transitions may have zero estimated probability. Sixth, this report uses a sample of 10,000 visitors and one chronological split. Finally, the holdout comparison uses one seed and 1,000 simulated journeys, so Monte Carlo variation is present. These limitations prevent interpreting the reported proportions as general population forecasts.

11. Future work

Future evaluation should use repeated visitor samples and multiple time-based splits, bootstrap or repeated-seed intervals, sensitivity analysis for inactivity thresholds, and calibration checks. Additional states or context features could be evaluated without obscuring the model's assumptions. A neural sequence model could be added as a separate comparator if the course objective requires deep learning, using the same carefully defined train/holdout protocol. External validation on another dataset would test whether the observed transition structure transfers.

12. Conclusion

This project builds an end-to-end, interpretable customer shopping behavior simulator from event logs. It cleans and samples visitor events, constructs sessions, maps events to behavioral states, estimates a first-order absorbing Markov chain, simulates journeys, and compares the resulting outcomes with a chronological holdout. For the documented 10,000-visitor sample, the observed purchase proportion was 0.6127% and the simulated proportion was 0.6000%. The result demonstrates a close aggregate match for this particular run, while the rare-purchase count and methodological assumptions call for cautious interpretation. The project is best presented as a reproducible probabilistic baseline rather than as a deep learning predictor.

References

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Appendix A. Reproducibility

The documented run used Python 3.12.3, 10,000 visitor identifiers, a 30-minute inactivity threshold, an 80/20 chronological session split, 1,000 simulated journeys, and random seed 42. The project output tables and charts record the run details. Re-running with the same input and settings is intended to make the random simulation reproducible within the configured software environment.

The dashboard can be started from the project folder with: .venv\\Scripts\\python.exe -m streamlit run app.py. The command-line real-data run uses: .venv\\Scripts\\python.exe run_project.py --sample-visitors 10000 --train-fraction 0.8 --journeys 1000 --seed 42.

Appendix B. Key terms

Term	Meaning in this study
Event row	One recorded visitor action such as a view or cart addition.
Visitor sample	Selected visitor identifiers; all available event rows for those visitors are retained.
Session	A sequence separated from the next by more than 30 minutes of inactivity.
Transition	A move between modeled states in adjacent event order.
Holdout	Later sessions excluded from fitting and used for descriptive comparison.
Simulated journey	A random state path generated using the learned transition matrix.